

Syed Jawad Ali

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DOB: 08 Dec 1992

Technical Skills

Programming: Python, JavaScript, TypeScript

Frameworks: Django, FastAPI, Next.js

Databases: PostgreSQL, MySQL, SQLite, pgvector

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Agentic AI and Multi-Agent Systems, Hallucination Mitigation

Retrieval & Vector Systems: ChromaDB, PostgreSQL Vector Store (pgvector), Hybrid Retrieval (Keyword + Vector), Embeddings, Similarity Search

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Reinforcement Learning (DQN)

Systems & Tools: Docker, Git, Redis, REST APIs, Scalable System Design

Publications

RAGENT: A Deep Q-Network-Driven Multi-Agent Framework for Adaptive Retrieval-Augmented Generation

Syed Jawad Ali, Dr.Sabina Akhar

Submitted to Springer (Under Review)

DQN-based multi-agent framework for adaptive retrieval in RAG systems, improving factuality, reducing hallucinations, and optimizing retrieval efficiency.

DQN-RAG: Deep Reinforcement Learning-Based Adaptive Retrieval for Multi-Hop Question Answering

Syed Jawad Ali, Dr.Arif Ur Rahman

Submitted to the International Conference on Digital Technologies and Transformation (ICoDT2 2026), hosted by NUST SEECS, Islamabad, Pakistan

Reinforcement learning-based RAG system that dynamically selects retrieval and reasoning actions, improving accuracy and reducing token usage on multi-hop QA tasks.

Experience

Joyn Digital

January 2023 – April 2026

Python Developer (Django, FastAPI, AI, GenAI – LLM/RAG)

Islamabad, Pakistan

- Designed and developed scalable backend systems using FastAPI and Django REST Framework, serving AI-powered applications.
- Built production-grade RAG (Retrieval-Augmented Generation) systems for intelligent document querying with citation support. Developed agentic AI pipelines integrating LLMs, enabling automated reasoning, task planning, and multi-step decision workflows.
- Integrated LLMs (BERT, Transformers, OpenAI APIs) with backend services to deliver real-time AI features. Implemented vector search systems (embeddings, similarity search) using PostgreSQL and optimized retrieval performance. Collaborated with cross-functional teams to deploy and scale AI services in production environments.

NextIN

April 2020 – May 2022

Backend and ML Developer

Islamabad, Pakistan

- Hands-on experience with machine learning libraries including Pandas, NumPy, Matplotlib, scikit-learn, Keras, and TensorFlow.
- Applied Machine Learning techniques including Supervised and Unsupervised algorithms for Classification, Clustering, and Dimensionality Reduction, as well as NLP using the Bidirectional Encoder Representations from Transformers (BERT) model and CNNs.

Zekab International

August 2016 – Dec 2018

Dot Net Developer

Islamabad, Pakistan

- Worked as ASP.NET Web Form and MVC developer.
- Transformed business logic into scalable backend solutions and collaborated with cross-functional teams to design, implement, and optimize systems.

Projects

DQN-RAG: Reinforcement Learning-Based Multi-Agent Retrieval System | *Django, LangGraph, VectorDB, PyTorch, Next.js*

- Developed a novel DQN-based adaptive Retrieval-Augmented Generation (RAG) framework that learns optimal retrieval actions based on query complexity and response quality feedback.
- Designed a LangGraph multi-agent system comprising planning, retrieval, web search, evaluation, and answer-generation agents for complex question answering tasks.
- Investigated reinforcement learning for retrieval optimization and hallucination mitigation through reward-driven agent coordination and iterative decision-making.
- Implemented a scalable production architecture using ChromaDB, PostgreSQL, Celery, Redis, Docker, and workflow automation for real-world deployment.

RAGENT: Intelligent AI Chatbot with Adaptive Retrieval | *Django, LangGraph, PostgreSQL, VectorDB, Next.js,*

- Developed a full-stack AI chatbot capable of dynamically selecting between knowledge retrieval and direct response generation based on query context and confidence signals.
- Implemented hybrid information access through document-based RAG and external web search, enabling accurate responses to both enterprise and real-time queries.
- Built a secure authentication and user management system featuring JWT authentication, OTP verification, password recovery, and role-based access control.
- Designed scalable REST APIs and integrated asynchronous processing to support responsive, production-ready conversational AI applications.

Multi-Agent AI System with Decision Intelligence (RAG + RL) | *Django, LangGraph, VectorDB, PostgreSQL, Next.js*

- Proposed and developed a multi-agent LLM framework consisting of Planner, Retrieval, and Response agents for complex query decomposition and execution.
- Designed and experimentally evaluated an RL-based decision module (state-action-reward formulation) for optimizing agent coordination, balancing accuracy, latency, and cost.
- Investigated adaptive reasoning strategies to reduce hallucination and improve response reliability through dynamic decision-making policies.

Intelligent RAG SaaS Platform (AI Knowledge Assistant) | *Django, LangGraph, VectorDB, PostgreSQL, Next.js*

- Designed and implemented a production-level Retrieval-Augmented Generation (RAG) system for enterprise knowledge retrieval over heterogeneous documents (PDF, CSV, Docs).
- Developed hybrid retrieval pipelines (keyword + vector) using PostgreSQL (pgvector), improving retrieval accuracy and grounding responses with citation-based generation.
- Explored techniques such as re-ranking, query rewriting, and confidence scoring to enhance factual consistency and reduce hallucination in LLM outputs.

AI-Enhanced Human Resource Management System | *Django, FastAPI, PostgreSQL, Next.js*

- Developed an HR analytics platform integrating machine learning models for salary prediction and performance evaluation.
- Applied data-driven modeling techniques to analyze employee attributes and improve decision-making processes in HR management.

Scalable E-commerce System with Intelligent Data Modeling | *Django, PostgreSQL, Next.js*

- Designed scalable backend architectures for large-scale transactional systems, focusing on efficient data modeling and system performance.
- Implemented advanced data management strategies for inventory, pricing, and recommendation-ready product structures.

Education

Bahria University

Islamabad, Pakistan

Master of Science in Computer Science (CGPA: 3.40)

2020

- **Relevant Coursework:** Data Science, Machine Learning, Deep Learning, and AI-based Thesis
- **Relevant Research:** Selection of Web Services using Artificial Intelligence

Federal Urdu University of Science and Arts

Islamabad, Pakistan

Bachelor of Science in Computer Science (CGPA: 3.01)

2016

- **Relevant Coursework:** Data Structures and Algorithms (C++), Programming Fundamentals (C++, Python), Web and Mobile Applications